INCH-POUND

MIL-DTL-52471/8B
4 December 2008
SUPERSEDING
MIL-DTL-52471/8A
27 April 1998

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLIES, RUBBER: HYDRAULIC PRESSURE-TYPE; TYPE 100RE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52471.

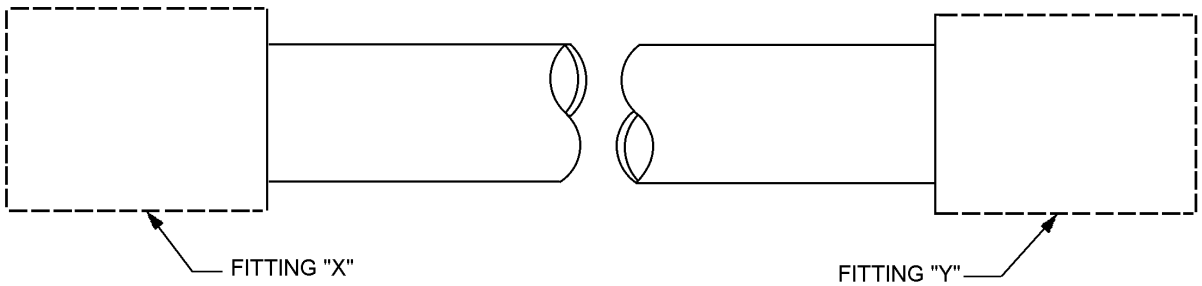


FIGURE 1. Hose assembly, fitting identification.

Operating pressure. Hose assembly operating pressures are shown in table I.

TABLE I. Hose operating pressure. 1/ 2/

Hose ID dash number	Hose ID			Maximum operating pressure		Minimum burst pressure	
	Inch fraction	Inch decimal	mm	psi	MPa	psi	MPa
-8	1/2	.500	12.70	7500	51.7	30,000	206.9
-12	3/4	.750	19.05	6250	43.1	25,000	172.4
-16	1	1.000	25.40	5000	34.5	20,000	137.9

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

Hose assembly code number. The purpose of this slash sheet is to provide hose assembly code numbers for determining the hose diameter, fitting X, fitting Y, and bend angles (when applicable) as shown in tables II through IV. The bend angles are called “style codes” and consist of hose assembly styles as explained in MIL-DTL-52471.

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TABLE II. 1/2-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN 1/	Fitting X (figure 1)		Fitting Y (figure 1)		Assembly code number 3/
		PIN	Bend angle 2/	PIN	Bend angle 2/	
1/2 .500 (12.70)	M52471/7-8	M52525/31-08-08	0°	M52525/31-08-08	0°	C001
				M52525/32-08-08	45°	C002
				M52525/33-08-08	90°	C003
		M52525/32-08-08	45°	M52525/32-08-08	45°	C004
				M52525/33-08-08	90°	C005
		M52525/33-08-08	90°	M52525/33-08-08	90°	C006
				M52525/34-08-08	0°	C007
		M52525/34-08-08	0°	M52525/35-08-08	45°	C008
				M52525/36-08-08	90°	C009
		M52525/35-08-08	45°	M52525/35-08-08	45°	C010
				M52525/36-08-08	90°	C011
		M52525/36-08-08	90°	M52525/36-08-08	90°	C012

1/ Part or Identifying Number (PIN).
2/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
3/ A suffix indicating plating finish shall be specified with the assembly code number, see table V.

TABLE III. 3/4-inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose PIN	Fitting X (figure 1)		Fitting Y (figure 1)		Assembly code number 2/
		PIN	Bend angle 1/	PIN	Bend angle 1/	
3/4 .750 (19.05)	M52471/7-12	M52525/31-12-12	0°	M52525/31-12-12	0°	C101
				M52525/32-12-12	45°	C102
				M52525/33-12-12	90°	C103
				M52525/34-12-12	0°	C107
		M52525/34-12-12	0°	M52525/35-12-12	45°	C108
				M52525/36-12-12	90°	C109
		M52525/35-12-12	45°	M52525/35-12-12	45°	C110
				M52525/36-12-12	90°	C111
		M52525/36-12-12	90°	M52525/36-12-12	90°	C112

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.
2/ A suffix indicating plating finish shall be specified with the assembly code number, see table V.

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TABLE IV. 1 inch hose assemblies.

Hose ID Inch fraction Inch decimal (mm)	Hose	Fitting X (figure 1)		Fitting Y (figure 1)		Assembly code number 2/
	PIN	PIN	Bend angle 1/	PIN	Bend angle 1/	
1 1.000 25.40	M52471/7-16	M52525/31-16-16	0°	M52525/31-16-16	0°	C201
				M52525/32-16-16	45°	C202
				M52525/33-16-16	90°	C203
				M52525/34-16-16	0°	C207
		M52525/34-16-16	0°	M52525/35-16-16	45°	C208
				M52525/36-16-16	90°	C209
		M52525/35-16-16	45°	M52525/35-16-16	45°	C210
				M52525/36-16-16	90°	C211
		M52525/36-16-16	90°	M52525/36-16-16	90°	C212

1/ Reference MIL-DTL-52471 for hose assembly bend angles and displacement angles used in the PIN.

2/ A suffix indicating plating finish shall be specified with the assembly code number, see table V.

Fitting materials and finishes shall be in accordance with MIL-DTL-52525, material and plating designators are specified in table V.

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TABLE V. Material and finish identification codes. 1/ 2/

PIN code material/plating finish	Material	Plating finish
A	Aluminum alloy 6061 or 7075	Anodize in accordance with MIL-A-8625, type II.
AN		Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type 2, class A.
BN		Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
C	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
CN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YC	Chrome-molybdenum steel alloy 4130	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
YN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
F	Steel	NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
FN	Chrome-molybdenum steel alloy 4130	
G	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 25.
YG	Chrome-molybdenum steel alloy 4130	
H	Steel	Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1. 3/
YH	Chrome-molybdenum steel alloy 4130	
J	Steel	Zinc plating with chromate conversion in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.
YJ	Chrome-molybdenum steel alloy 4130	
K	Nickel-copper alloy UNS N04400	No additional finish.
M	High-chromium nickel alloy UNS N06690	No additional finish.
N	Steel	Zinc aluminum in accordance with ASTM F1136, grade 3, NC.
YN	Chrome-molybdenum steel alloy 4130	
P	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 5.
YP	Chrome-molybdenum steel alloy 4130	
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.
SN		Passivation above and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
T	Titanium	Anodize in accordance with SAE-AMS2488, type 2.
V	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 1.
W	Steel	Any zinc plating above.
YW	Chrome-molybdenum steel alloy 4130	
Z	Steel	Any zinc plating above with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YZ	Chrome-molybdenum steel alloy 4130	

1/ All materials and finishes shall be in accordance with MIL-DTL-52525.
2/ Embitterment test need not be run.
3/ Zinc phosphate finish is hexavalent chromium free.

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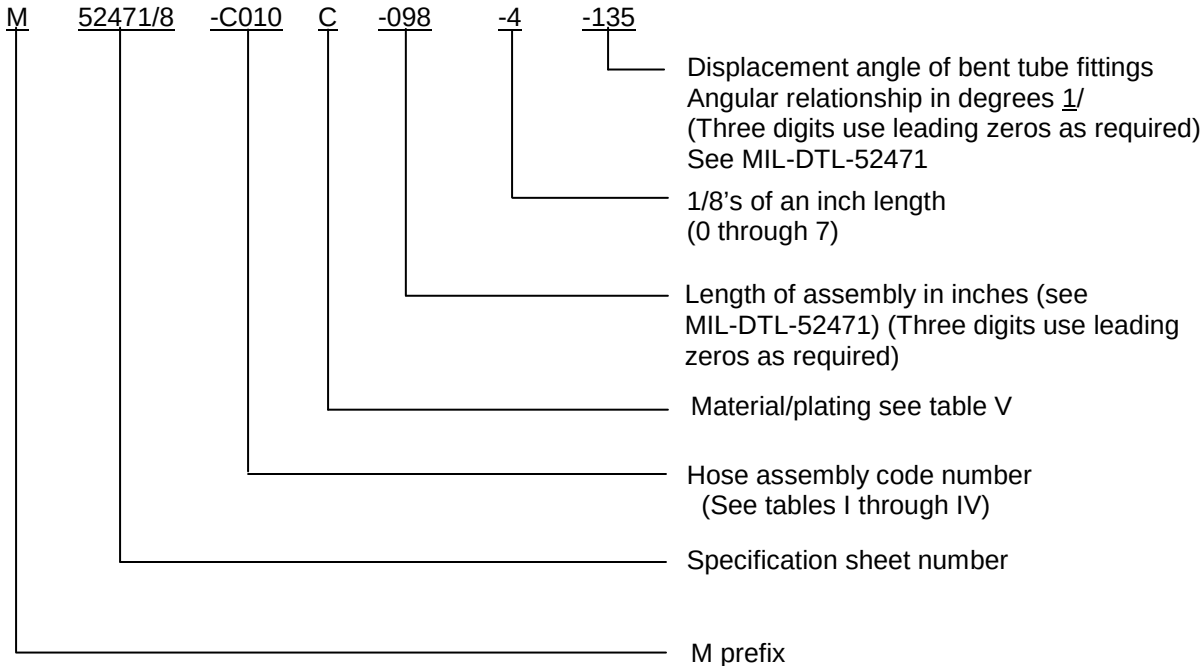
REQUIREMENTS:

The hose assemblies shall be as specified on figure 1 and in tables I through V.

Hose shall be in accordance with MIL-DTL-52471/7.

Connector fittings for hose assemblies shall be in accordance with MIL-DTL-52525.

PIN: The PIN consists of the letter “M”, the specification sheet number, an assigned code number for the hose assembly, material/plating designator, hose assembly length, and angular relationship of fittings, when applicable.



1/ Not applicable for assemblies with straight fittings.

PIN example: M52471/8-C010C-098-4-135, describes a hose assembly consisting of M52471/7-8 hose, fitting X M52525/35-08-08, bend angle 45°, fitting Y M52525/35-08-08, bend angle 45°, steel with cadmium plating, 98 ½ inches long, and a displacement angle of 135°.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-52471, this document references the following:

MIL-A-8625	ASTM B633	SAE-AMS2417
MIL-DTL-16232	ASTM B695	SAE-AMS2488
MIL-DTL-52471/7	ASTM F1136	SAE-AMS2700
MIL-DTL-52525	SAE-AMS-C-81562	
MIL-DTL-81706	SAE-AMS-QQ-P-416	

CONCLUDING MATERIAL

Custodians:
Army - CR4
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2007-024)

Review activities:
Army - GL
Navy - CG, MC, SA

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